Sahaj Bhadja

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Summary

Full-stack Software Engineer with 2+ years shipping AI-native education products, from **React/TypeScript** interfaces to **Python** services and **LLM** grading, feedback, and retrieval workflows, with CI/CD and observability.

Education

University of Southern California, Los Angeles, CA **August 2024 – May 2026**
Master of Science in Computer Science *GPA: 3.75/4*
Coursework: Advanced NLP, Web Technologies, Database Systems, Analysis of Algorithms

Pandit Deendayal Energy University, Gandhinagar, India **November 2020 – May 2024**
Bachelor of Technology in Computer Science and Engineering *GPA: 3.88/4*
Coursework: Software Engineering, Data Retrieval, Object-Oriented Programming, Operating Systems

Experience

USC Dornsife **September 2025 – Present**
Software Engineer *Los Angeles, CA*
- Architected a multi-tenant **AI education platform** on Azure SQL and Blob Storage with row-level security for 500+ users, cutting dashboard latency from 1.2s to 350ms via query and index tuning.
- Restructured fault-tolerant async **LLM orchestration** with FastAPI workers and explicit job-state transitions, processing 1K+ AI jobs at 3x throughput while CI/CD-backed data pipelines sustained 99%+ production reliability.

Grad8 **June 2025 – August 2025**
Software Engineer Intern *Los Angeles, CA*
- Built an AI grading product with **React** and Flask where rubric generation triggers on reference upload and multimodal LLMs score answers with **automated feedback**, serving 150+ instructors and 10K+ daily requests.
- Designed **prompt chains** across GPT-4o, Claude, and Qwen-VL with OCR fallback and per-call token cost tracking, achieving 90%+ grading consistency and cutting feedback turnaround 60% for instructors.

Aasma Technology Solutions **August 2023 – July 2024**
Software Developer Engineer – SecureSync *India*
- Owned cross-platform data-sharing features from requirement analysis through deployment on a microservice architecture, building 20+ unit and integration test suites with **CI/CD** in Agile sprints with Microsoft stakeholders.
- Optimized APIs for low-bandwidth users with pagination, field filtering, and caching, improving response times 30%, and led an **ETL** redesign on AWS Glue and S3 boosting pipeline efficiency 68%.

Aasma Technology Solutions **May 2023 – August 2023**
Full Stack Developer Intern – VibrantChem *India*
- Built a CRM platform with **React**, **TypeScript**, Node.js, and MongoDB, delivering real-time WebSocket updates across eight microservices for resources, billing, leads, and notifications while syncing 2M+ records.
- Used Sentry, Azure Monitor, and Kibana for **observability** and Google Analytics for feature monitoring, turning usage insights into changes that cut resource wastage 42% and raised lead conversions 15%.

LST Capitals **November 2022 – January 2023**
Web Developer Intern *India*
- Gathered **client requirements**, designed the end-to-end architecture across 12+ integration points, and presented it for sign-off before building Python Flask APIs behind React Native and React TypeScript clients.
- Shipped booking, pricing, coupon, and payment APIs with Firebase **real-time** sync and hourly Databricks Spark analytics over 50K+ records, sustaining sub-200ms latency and 40% faster booking confirmations.

Skills

Languages: TypeScript, JavaScript, Python, Java, SQL, HTML, CSS
Frontend: React, Next.js, TailwindCSS, Shadcn, Responsive UI
Backend & Data: FastAPI, Flask, Node.js, REST APIs, WebSockets, PostgreSQL, Supabase, MySQL, MongoDB
AI/ML: LLM Integration, Prompt Chaining, Embeddings, RAG, FAISS, Automated Evaluation, OCR, PyTorch, Transformers
Cloud & DevOps: AWS, Microsoft Azure, GCP, Docker, GitHub Actions, CI/CD, Vercel
Practices: Code Reviews, Unit and Integration Testing, Observability, Sentry, Requirements Gathering, Agile/Scrum, Cost Monitoring

Projects

Big-Brain Live | *LLaMA-2, FAISS, Embeddings, Clerk, Convex, PostgreSQL*
- Built a document Q&A and summarization app using LLaMA-2 with transformer **embeddings** and FAISS retrieval over 50K+ documents, reaching 95% query accuracy for 200+ users with Clerk auth and Convex collaboration.

CaptionCraft Live | *PyTorch, Transformers, BLIP, GIT, ViT-GPT-2, Flask*
- Benchmarked BLIP, GIT, and ViT-GPT-2 against a custom model on 10K images and 100K captions, using **evaluation** results to lift caption accuracy from 81% to 96% behind real-time inference APIs.

PiXMod Live | *Next.js, CLIP, SAM, Stable Diffusion, MongoDB, Stripe*
- Built an AI SaaS image platform in Next.js with SAM segmentation, Stable Diffusion inpainting, and CLIP **embedding** search across 100K+ images, adding Stripe subscriptions that raised premium conversion 18%.